

# Study design, cohort qualification, and technical consistency

Frozen centroids, explicit cohort roles, and projection-confidence summaries define the validation engine.

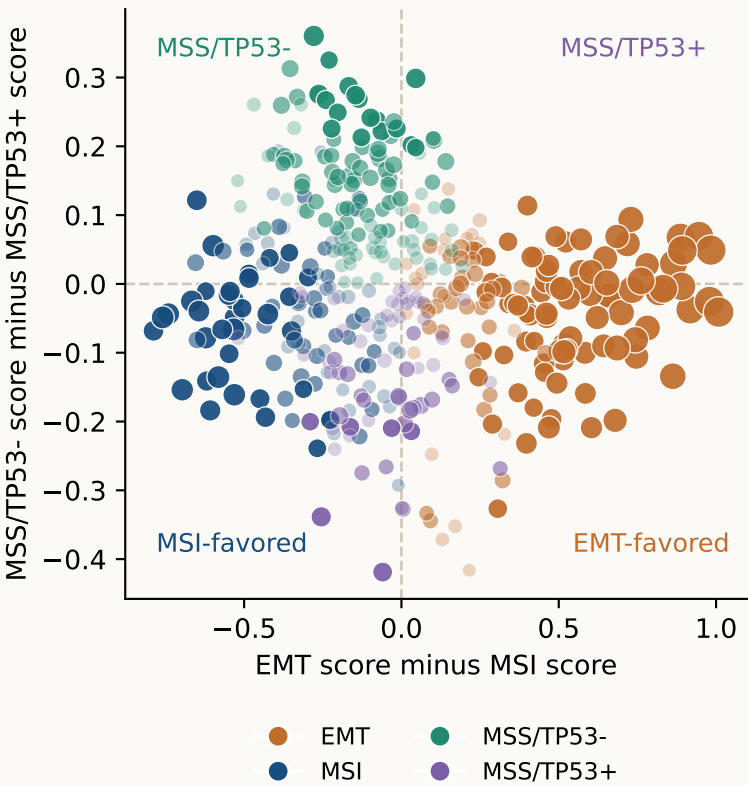
## A Frozen derivation-to-validation workflow



## B Cohort roles and endpoint context

| Cohort    | Role         | N        |
|-----------|--------------|----------|
| GSE62254  | Derivation   | 300      |
| GSE15459  | Replication  | 192      |
| GSE84437  | Replication  | 433 used |
| TCGA-STAD | Orthogonal   | 443      |
| GSE26899  | Tx extension | 93       |
| GSE26901  | Tx extension | 109      |

## C Centroid space



## D Projection QC

